Reflecting on Decision-Making and Problem-Solving

Scenario One: Choosing an English-Medium Mathematics Degree

My original decision-making process

One of the most important decisions I have made concerned the language of my first bachelor's degree in Mathematics. I had completed my school education in Azerbaijani and was initially admitted to an Azerbaijani-medium university group. Shortly afterwards, I was offered the opportunity to transfer to a newly created experimental English-medium Mathematics group. It was the first group of its kind in the university's history, so there was little previous experience from which I could judge how successful it would be.

I was given approximately three days to make the decision. Remaining in the Azerbaijani-medium group appeared to be the safer option. I already understood mathematical terminology in Azerbaijani and could expect to concentrate primarily on the subject rather than simultaneously learning through a second language. Transferring involved considerable uncertainty because every lecture, textbook, examination and classroom discussion would be in English. Although I enjoyed English, I had never previously studied all my subjects through it. I worried that I might misunderstand lectures, receive lower grades or struggle to express mathematical ideas accurately.

Nevertheless, I felt strongly attracted to the English-medium programme. I loved learning English and imagined that combining it with Mathematics would broaden my future opportunities. I believed it might allow me to access international materials, communicate with a wider academic community and eventually work in an international environment. I did not construct a formal comparison of the two programmes, investigate employment statistics or calculate the consequences of each option. My decision was largely intuitive and based on enthusiasm, hope and a general impression that English would be valuable in my future.

After thinking for three days, I transferred to the English-medium group. The first period was challenging because I had to learn mathematical content and unfamiliar terminology simultaneously. However, the decision eventually became one of the most beneficial choices of my life. I developed the ability to study and teach Mathematics in English and later obtained employment in international schools. This increased both my professional opportunities and earning potential. Our experimental group also proved successful: in later years, applicants could select the English-medium Mathematics programme directly rather than entering another group and transferring later.

Reconsidering the decision using module knowledge

Although the outcome was positive, this does not necessarily mean that my original process was optimal. Knowing the result may create hindsight bias, making the English-medium option now appear more obviously correct than it actually was at the time. The programme was experimental, my ability to manage the language demands was uncertain, and success was not guaranteed. Halpern and Dunn (2023) emphasise that effective decision-making requires deliberate consideration of alternatives, consequences, evidence and possible cognitive biases rather than relying solely on an appealing intuition.

If I could reconsider the decision, I would first frame it more carefully. My original framing was approximately, “Should I remain in the comfortable Azerbaijani group or take the risk of studying in English?” This frame emphasised comfort against difficulty. A more useful question would be: “Which programme best supports my long-term academic, professional and personal goals while presenting a manageable level of risk?” Decision frames influence which alternatives and consequences become psychologically salient, even when the underlying facts remain unchanged (Tversky & Kahneman, 1981).

I would then collect information that I did not seek originally. I would ask who would teach the programme, whether language support would be available, which textbooks would be used, whether transferring back was possible, and how the university would support an experimental group. I would also speak to students who had previously studied individual subjects in English and complete sample readings or lectures to assess my readiness.

Next, I would apply a weighted utility procedure. For example, I might give long-term career opportunities a weight of .30, personal interest .20, English development .20, academic risk .20 and immediate comfort .10. Using a five-point scale, I might retrospectively rate the English programme as 5, 5, 5, 2 and 1 respectively, producing:

$$(.30 \times 5) + (.20 \times 5) + (.20 \times 5) + (.20 \times 2) + (.10 \times 1) = 4.0$$

The Azerbaijani programme might receive 3, 3, 1, 5 and 5:

$$(.30 \times 3) + (.20 \times 3) + (.20 \times 1) + (.20 \times 5) + (.10 \times 5) = 3.2$$

These scores would not be objective truths, but they would make my priorities and assumptions visible. I could then test how much the result changed when different weights were used.

I would also actively check for optimism bias and confirmation bias. Because I loved English, I may have focused on evidence supporting the transfer while underestimating academic difficulties. To counter this, I would deliberately list reasons not to transfer and ask what evidence would show that my preferred choice was unwise. Metacognitive monitoring, examining the quality of one’s own reasoning is central to Halpern’s approach to critical thinking and transfer (Halpern, 1998).

The module-informed process would probably have produced the same decision, but for stronger reasons. My original approach was intuitive, emotionally positive and relatively narrow. The revised approach would be transparent, evidence-based and sensitive to uncertainty. It would not guarantee success, but it would allow me to explain why the potential long-term benefits justified the short-term academic risk. The main lesson is that a good outcome should not be confused with a flawless decision process.

Scenario Two: Solving an Unsustainable Employment Problem

My original problem-solving process

A separate significant problem developed during my employment at my first international school. I worked there for eight years and had strong relationships with my students and colleagues. The school was also close to my home, which made daily travel convenient. However, my workload gradually became excessive. I regularly took work home and continued completing tasks during weekends. I felt that my salary did not adequately reflect the time, responsibility and effort required by the role.

Initially, I tried to solve the problem within the school. I approached the management and requested a salary increase. I explained that my workload was substantial and that my compensation no longer seemed proportionate to my responsibilities. The management did not approve the request. After this rejection, I felt disappointed because I had invested many years in the school and cared deeply about its community.

I did not initially create a formal problem-solving plan. For some time, I continued working under the same conditions while thinking about whether I should stay. Leaving involved emotional and practical risks. I would lose daily contact with familiar colleagues and students, give up a short commute and enter an unfamiliar workplace. Nevertheless, the workload and lack of recognition were becoming increasingly difficult to accept.

Eventually, I began exploring positions in other international schools. I found a school offering higher pay, a more manageable workload and a culture that appeared to value its staff more strongly. I prepared for and completed its recruitment examination, was accepted and left my previous school. I have now worked in the new school for four years. I no longer routinely take work home, and the improved balance and working conditions have confirmed that changing schools was an effective solution.

Reconsidering the problem using module knowledge

From a module-informed perspective, I would begin by constructing the problem space more explicitly. Newell and Simon (1972) conceptualised problem-solving as a search through a mentally represented space containing a current state, a goal state, available operators and constraints.

My initial state consisted of excessive workload, inadequate compensation, weekend work and declining satisfaction. The goal state should not simply be “receive a salary increase” or “leave the school.” A better goal would be a sustainable teaching role offering fair compensation, reasonable workload, professional respect and an acceptable commute.

Relevant objects and information would include my actual weekly working hours, the amount of unpaid work completed at home, contractual responsibilities, salary levels in comparable schools, available vacancies, recruitment requirements and the likely effect of changing schools on my personal life. Possible operators would include negotiating pay, requesting reduced duties, seeking administrative support, changing role within the school, applying elsewhere or completing additional qualifications. Constraints would include financial security, contractual notice requirements, loyalty to students, commuting distance and the need to obtain another position before resigning.

My original representation may have been too narrow because I initially treated a salary increase as the principal solution. When management refused, the problem could have appeared temporarily unsolvable. I would now apply means–ends analysis, comparing the initial state with the broader goal and creating subgoals that reduced the difference. These would be:

1. Record my real workload for several weeks.
2. Compare my responsibilities and salary with similar positions.
3. Prepare a specific proposal involving pay, workload and role boundaries.
4. Identify possible internal solutions.
5. Research alternative schools.
6. Update my CV and prepare for recruitment assessments.
7. Secure and evaluate an offer before resigning.

I could also use working backwards. If the goal were to begin the next academic year in a sustainable role, I would work backwards from signing a contract, receiving an offer, passing the examination, submitting an application and identifying suitable vacancies. This would turn a stressful, emotionally complex problem into a sequence of manageable actions.

An important obstacle was my eight-year attachment to the school. My past investment was personally meaningful, but those eight years could not be recovered regardless of whether I stayed or left. Continuing merely because I had already invested substantial time would resemble sunk-cost reasoning. I also had to restructure the problem from “How can I make this school pay me more?” to “How can I obtain sustainable and fairly valued employment?” This representational change expanded the problem space and revealed solutions outside the existing institution.

The Hill-Briggs model further suggests that effective problem-solving depends on general strategies, transfer of previous experience, domain-specific knowledge, an adaptive problem-solving orientation and the surrounding environment (Hill-Briggs, 2003). The model was developed as an evidence-based account of problem-solving in chronic illness self-management, but its interacting components can also illuminate broader practical problems. My teaching expertise strengthened my application, my prior experience helped me identify desirable and undesirable workplace conditions, and persistence enabled me to complete the recruitment process despite uncertainty. External factors—available vacancies, management practices and support from other people—also shaped which solutions were realistic.

My original solution was ultimately successful, but the route towards it was reactive. I first requested more money, experienced rejection and only then widened my search. A module-informed approach would have defined the problem more broadly from the beginning, gathered objective evidence and generated several alternatives before acting. It might have reduced the period during which I continued tolerating unsustainable conditions. However, it would not have removed all uncertainty: a new school can appear attractive before employment but prove different in practice. The improved process would therefore include ongoing metacognitive monitoring and a later review of whether the new role genuinely met the criteria established at the outset.

Conclusion

These scenarios demonstrate that successful outcomes can emerge from relatively intuitive and informal processes. However, module-informed thinking would have made both processes more systematic and defensible. In the university decision, structured framing, information gathering, utility weighting and bias checking would have clarified why the long-term opportunity justified the risk. In the employment problem, constructing a fuller problem space, creating subgoals and restructuring the goal would have revealed alternatives earlier. The principal improvement is not certainty, because neither major decisions nor real-world problems permit complete certainty. Rather, it is the ability to make assumptions explicit, evaluate evidence, monitor reasoning and revise an approach when it is no longer effective.

References

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